

本说明分为中文和英文两部分，其中前面为中文说明，后面为英文说明。

This instruction is divided into two parts, Chinese and English. The Chinese description is in the front and the English description is at the back.

目录 Contents

1 实时数据	2
2 设备信息	3
3 高级设置	4
3.1 EG1615 参数修改	4
3.2 EG8026/EG8025 参数修改	5
4 固件升级	6
5 控制台	7
5.1 EG1615 控制台	7
5.2-EG8026 控制台	8
5.2.1-EG8026 控制台使用说明	8
5.2.1-EG8025 控制台使用说明	9
6 Real-Time Data	9
7 Device Information	10
8 Setting	11
8.1 EG1615 Parameter Modification	11
8.2 EG8026/EG8025 Parameter Modification	12
9 Update	13
10 Console	14
10.1 EG1615 Console	14
10.2-EG8026 Console	16
10.2.1-EG8026 Console Usage Instructions	16
10.2.1-EG8025 Console Usage Instructions	17
11 Q&A	18
11.1 Q: 如何校准电池电压? How to calibrate the battery voltage?	18
11.2 Q: 如何校准 AC 输出电压和输出电流? How to Calibrate the AC Output Voltage and Output Current?	18
11.3 Q: 如何修改参数? How to modify the parameters?	18
11.4 Q: 如何校准电池电流? How to calibrate the battery current?	18
11.5 Q: 如何校准 EG8025 的功率或电流? How to calibrate the power or current of EG8025?	19

1 实时数据

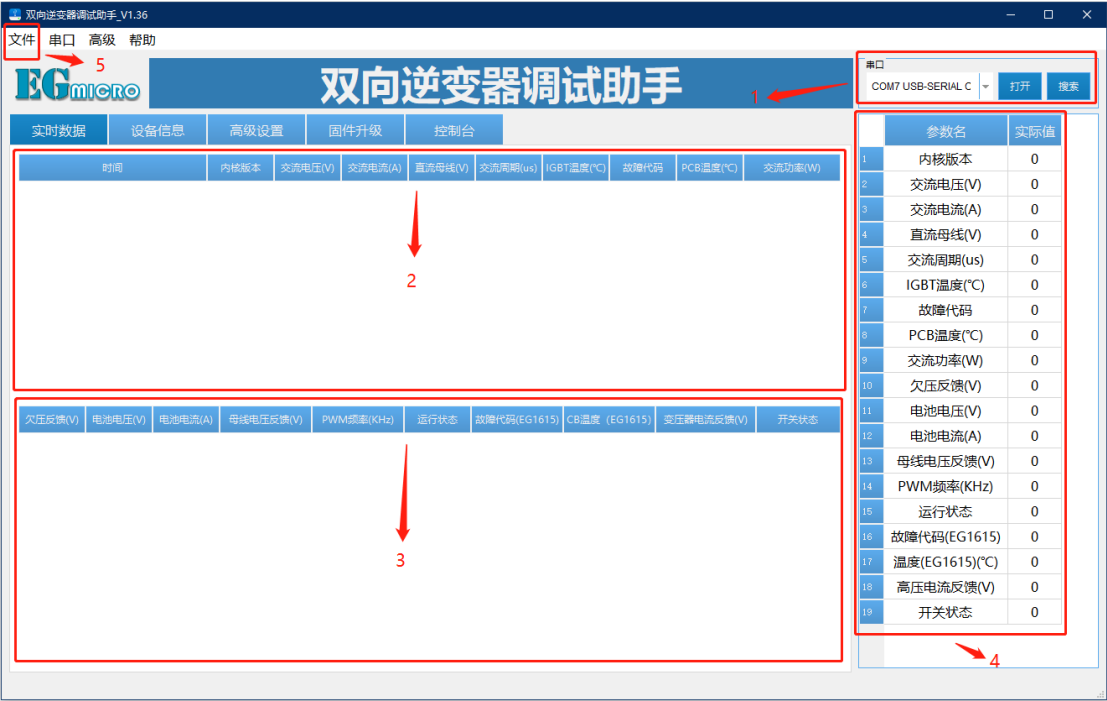


图 1-1

1. 连接串口：点击搜索后，选择 CH340 设备，再点打开。
2. 逆变部分数据显示
3. 电池部分数据显示
4. 全部数据显示:温度>40°C或≤0°C会标红，故障码异常会标红。
5. 导出数据：打开“文件”按钮，选择“导出到 excel”，可将 2 和 3 的数据导出；如遇到测试问题，将导出的数据与问题现象发送给工程师分析。

2 设备信息



图 2-1

- 1 读 DID 配置：点击“读”读取单条 DID 配置，点击“读全部”读取全部 DID 配置。
- 2 输入待写入值
- 3 写 DID 配置：点击“写”将待设置的值写入，写入即生效。
- 4 开环输出，闭环输出：低压 IO 测试使用，正常工作禁止使用。

3 高级设置

3.1 EG1615 参数修改



图 3-1-1

- 1 总电量设定和 SOC：保留
- 2 读 CFG 配置：点击“读”读取单条 CFG 配置，点击“读全部”读取全部 CFG 配置。
- 3 写 CFG 配置：点击“写”将对应寄存器中待设置的值写入，确保参数完整，该操纵不保存，掉电会恢复。
- 4 保存所有：点击“保存所有”按钮后，将所有参数保存。
- 5 恢复默认：点击“恢复默认”按钮后，高级设置参数恢复成出厂设置，设备信息参数不恢复。
- PS：恢复默认后等待 5S 后重新上电

3.2 EG8026/EG8025 参数修改



图 3-2-1



图 3-2-2

PS: EG8026 执行下面 1~4 步骤后需要将 EG1615 高级设置-逆变控制参数写入 0（即自定义模式）才会运行修改后的参数。EG8025 执行下面 1~4 步骤即可

- 1 读 CFG 配置：如图 3-2-1，点击“读全部”读取全部 CFG 配置。
- 2 保存所有：如图 3-2-1，点击“保存所有”按钮后，将所有参数保存到自定义区。
- 3 切换到自定义区：如图 3-2-2，将逆变器配置待设置填入 0 后，点写入，即可切换到自定义区。PS：逆变器配置 0 为自定义区，1 为默认参数
- 4 修改参数：如图 3-2-1，点击“写”将对应寄存器中待设置的值写入，确保参数完整，该操纵不保存，掉电会恢复。
点击“保存所有”按钮后，所修改的参数才会保存。
- 5 恢复默认：如图 3-2-1，点击“恢复默认”按钮，所有信息会恢复到出厂设置，包括设备信息的参数。
PS：恢复默认后等待 5S 后重新上电

4 固件升级

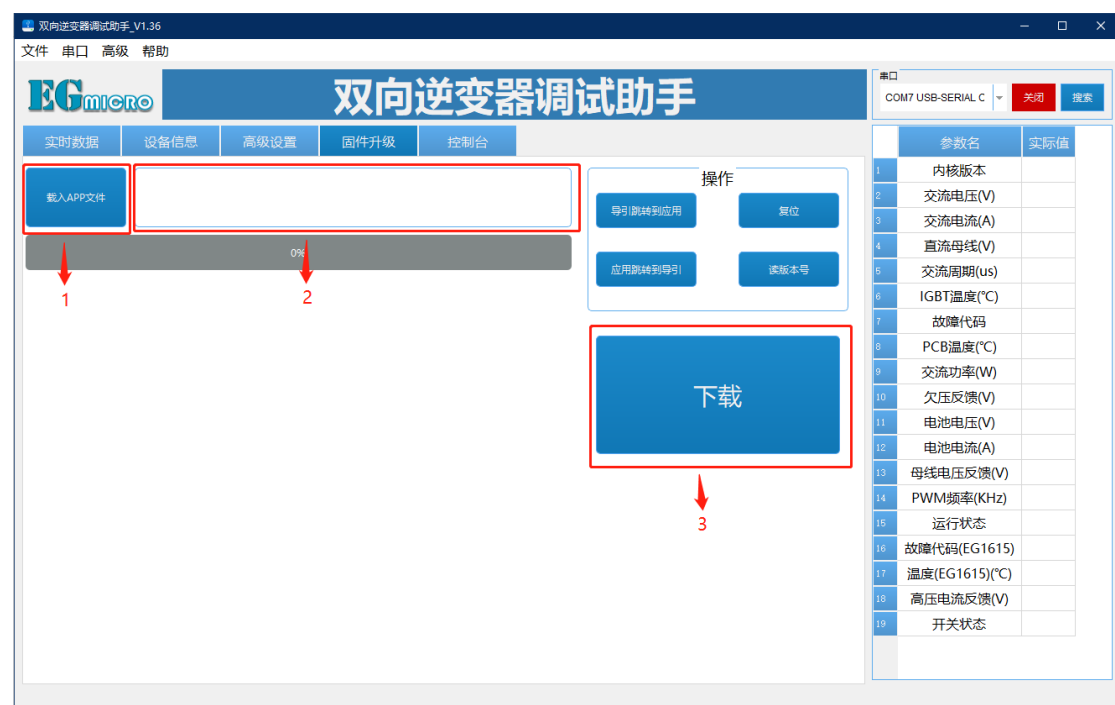


图 4-1

- 1 载入 APP 文件：点击“载入 APP 文件”，选择文件
- 2 显示文件路径：文件也可直接拖入
- 3 更新程序：点击“下载”按钮，进入下载程序模式。

PS：

- ① 如遇到下载出错，直接再点击“下载”按钮即可。
- ② 如果整机在工作状态时，串口接在 EG1615 的情况下，如需要更新 EG8026 的程序，需要先从 EG1615 透传到 EG8026 后，再点击下载，下载完成后需要透传回 EG1615。
(透传功能请看控制台说明)
- ③ 如果单独更新驱动板程序，串口接在 EG1615 的情况下，直接装载 EG1615 的 APP 文件，串口接在 EG8026 的情况下，直接装载 EG8026 的 APP 文件。

5 控制台

5.1 EG1615 控制台

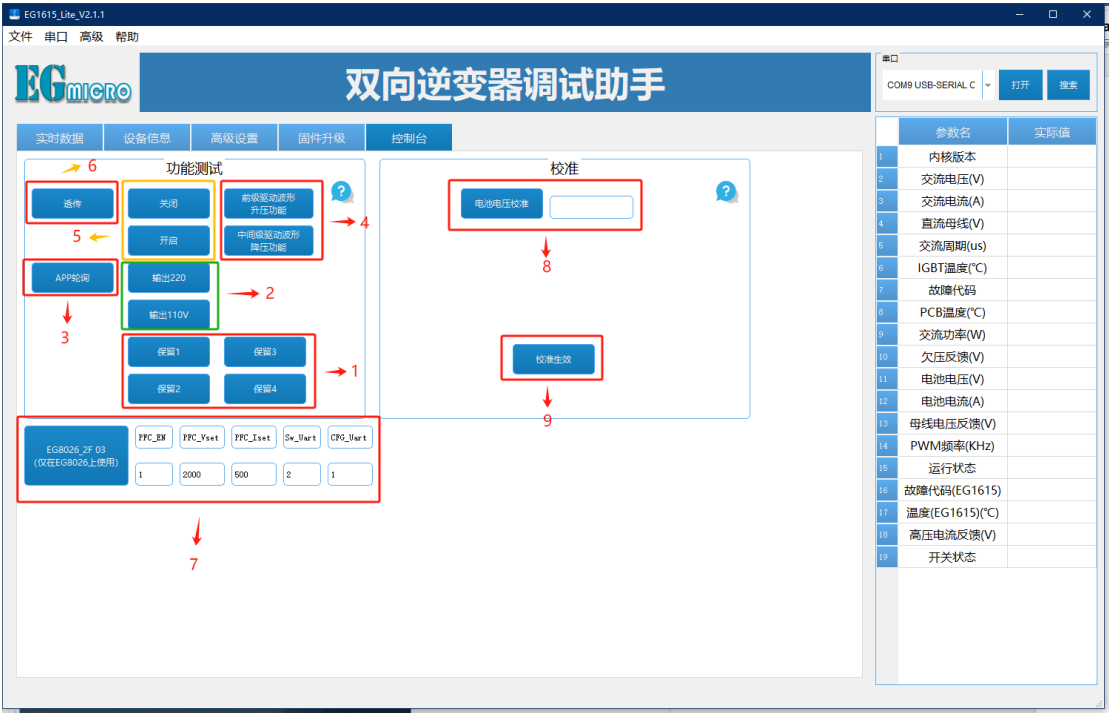


图 5-1

PS:当串口接在 EG1615 上使用说明

- 1 四个按钮为调试用，没有实际功能
- 2 输出 220V：EG8026 切换到配置 1；输出 110V：EG8026 切换到配置 2。实际输出以 EG8026 的配置为准。
- 3 APP 轮询：当 session 为 3 时，APP 报文停止，可以通过该按钮获取 APP 数据。
- 4 升压测试，降压测试：该按钮为测试按钮，仅在低压 IO 测试使用，正常工作禁止使用。
- 5 开启：放电状态为开启逆变，充电状态为开启充电；
关闭：放电状态为关闭逆变，充电状态为停止充电。
- 6 透传：意思是通过 EG1615，接收和操作 EG8026 的指令。
具体操作：串口接在 EG1615 时，点击“透传”按钮后，可以读取 EG8026 的设备信息和高级设置参数，同时也可以使用 EG8026 的校准和修改参数等功能。再次点击“透传”按钮后，可以切回到 EG1615，正常工作状态串口需要读取 EG1615 的数据，否则会出现逆变不输出或充电不充的情况。
- 7 EG8026_2F_03 按钮：该按钮为测试 EG8026 的 PFC 功能，正常工作禁止使用。
- 8 校准电池电压：在正常工作状态下，量取实际电池的电压值*100 后，保留整数部分后填入右侧输入框中，再点击“校准电池电压”按钮，此时校准是不保存的。
- 9 校准生效：确保校准正确后，点击“校准生效”按钮，此时校准值保存。

PS：所有校准需要在逆变状态

5.2•EG8026 控制台

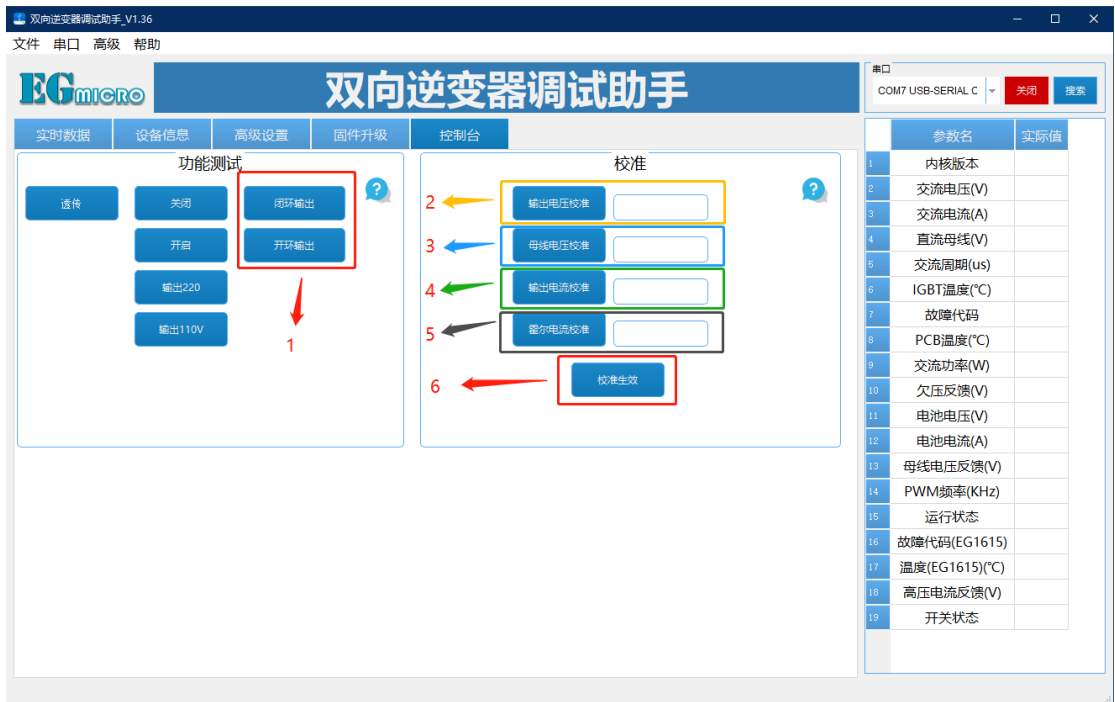


图 5-2

5.2.1•EG8026 控制台使用说明

PS 当串口接在 EG8026 上使用说明（所有校准需要在逆变状态，电流需要在满载时校准）

- 1 闭环输出：控制 EG8026 闭环输出；开环输出：控制 EG8026 开环输出；
- 2 输出电压校准：在正常工作状态下，量取实际 AC 电压的电压值*10 后，保留整数部分后填入右侧输入框中，再点击“输出电压校准”按钮，此时校准是不保存的。
- 3 母线电压校准：在正常工作状态下，量取实际母线电压的电压值*10 后，保留整数部分后填入右侧输入框中，再点击“母线电压校准”按钮，此时校准是不保存的。
- 4 输出电流校准：（该校准值对应 H 桥臂下的采样电阻总电流）在正常工作状态下，量取实际 AC 电流的电流值*100 后，保留整数部分后填入右侧输入框中，再点击“输出电流校准”按钮，此时校准是不保存的。
- 5 霍尔电流校准：（该校准值对应霍尔芯片采集的电流）在正常工作状态下，量取实际 AC 电流的电流值*100 后，保留整数部分后填入右侧输入框中，再点击“霍尔电流校准”按钮，此时校准是不保存的。
- 6 校准生效：确保校准正确后，点击“校准生效”按钮，此时所有校准值保存。

PS.其余按钮请看 EG1615 控制台说明。

5.2.1•EG8025 控制台使用说明

PS2:当串口接在 EG8025 上使用说明（所有校准需要在逆变状态，电流需要在满载时校准）

- 1 闭环输出：控制 EG8025 闭环输出；开环输出：控制 EG8025 开环输出；
- 2 输出电压校准：在正常工作状态下，量取实际 AC 电压的电压值*10 后，保留整数部分后填入右侧输入框中，再点击“输出电压校准”按钮，此时校准是不保存的。
- 3 母线电压校准：在正常工作状态下，量取实际母线电压的电压值*10 后，保留整数部分后填入右侧输入框中，再点击“母线电压校准”按钮，此时校准是不保存的。
- 4 输出电流校准（该校准值对应 EG8025 中 H 桥左桥臂的采样电阻电流）：EG8025 不建议使用该校准方法
- 5 霍尔电流校准（该校准值对应 EG8025 中 H 桥右桥臂的采样电阻电流）：EG8025 不建议使用该校准方法
- 6 校准生效：确保校准正确后，点击“校准生效”按钮，此时所有校准值保存。

PS.其余按钮请看 EG1615 控制台说明。

6 Real-Time Data

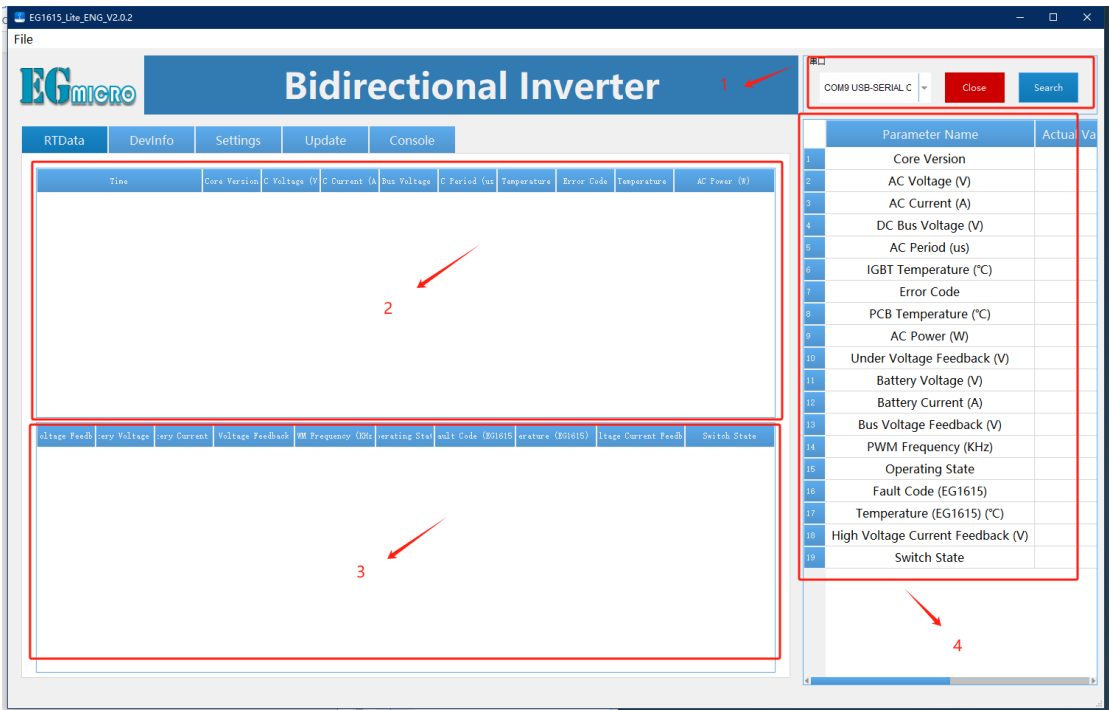


Figure 6-1

1. **Connect the serial port:** After clicking "Search", select the CH340 device, and then click "Open".
2. **Inverter part data display**

3. **Battery part** data display
4. **All data display:** Temperatures $> 40^{\circ}\text{C}$ or $\leq 0^{\circ}\text{C}$ will be marked red, and abnormal fault codes will be marked red.
5. **Export data:** Click the "File" button and select "Export to Excel" to export the data in items 2 and 3. In case of testing problems, send the exported data and the problem phenomena to the engineer for analysis.

7 Device Information

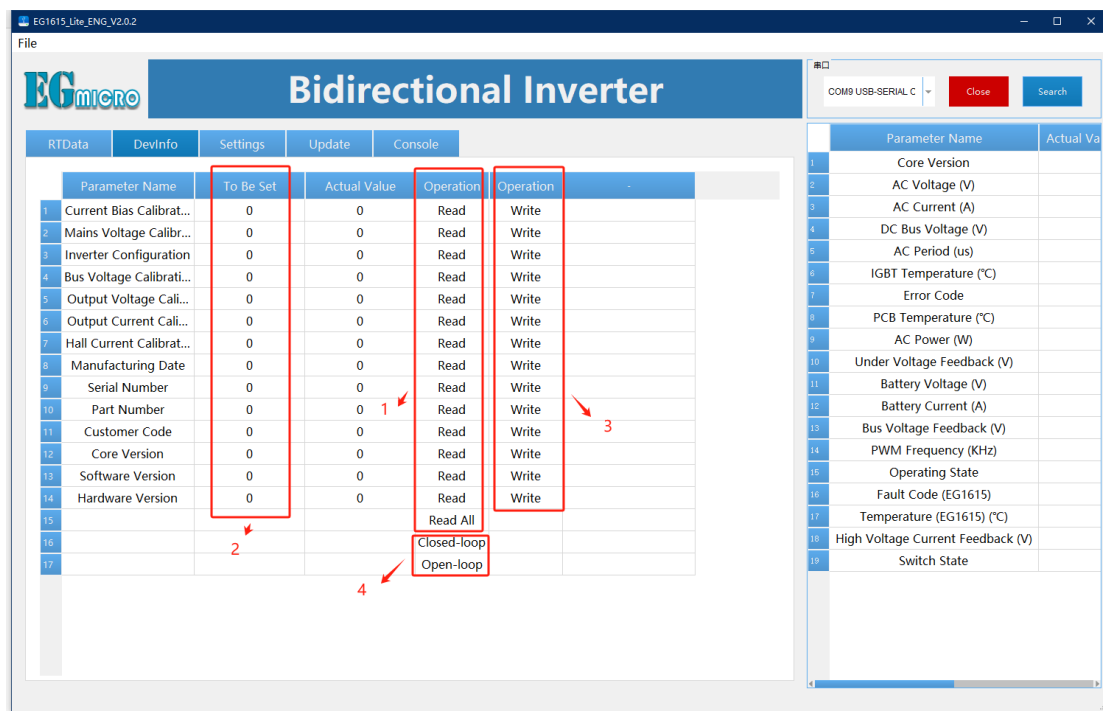


Figure 7-1

1. **Read DID configuration:** Click "Read" to read a single DID configuration, and click "Read All" to read all DID configurations.
2. Enter the value to be written.
3. **Write DID configuration:** Click "Write" to write the value to be set, and the setting takes effect immediately upon writing.
4. **Open-loop output, closed-loop output:** For low-voltage IO testing. Prohibited during normal operation.

8.2 EG8026/EG8025 Parameter Modification

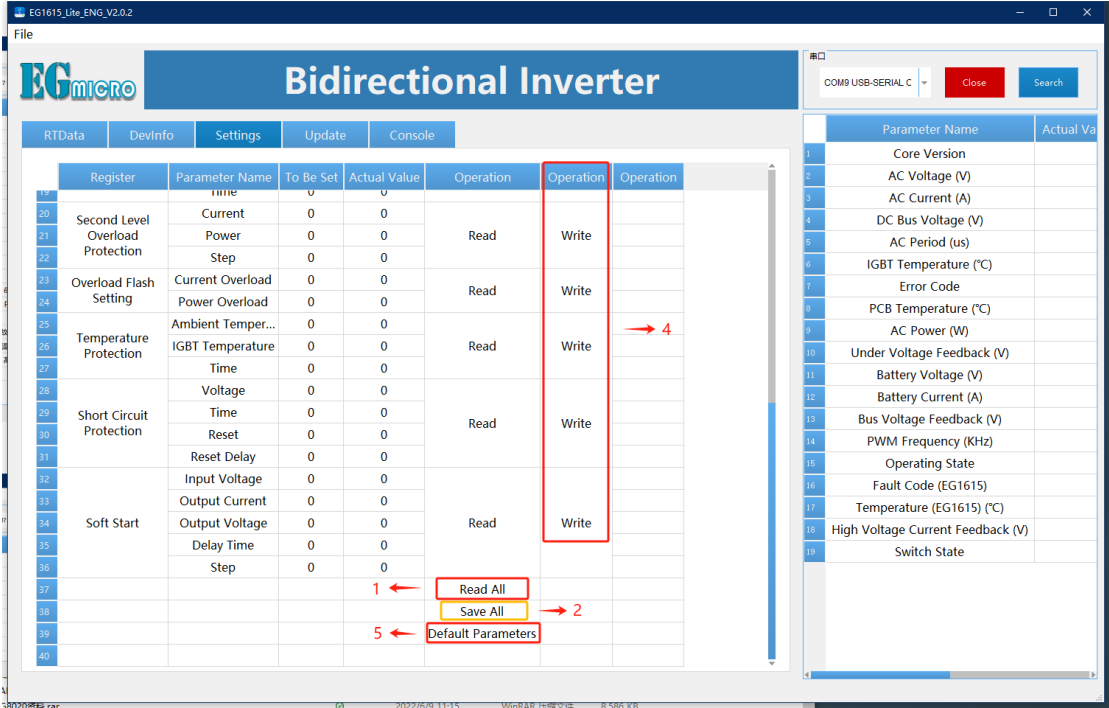


Figure 8-2-1

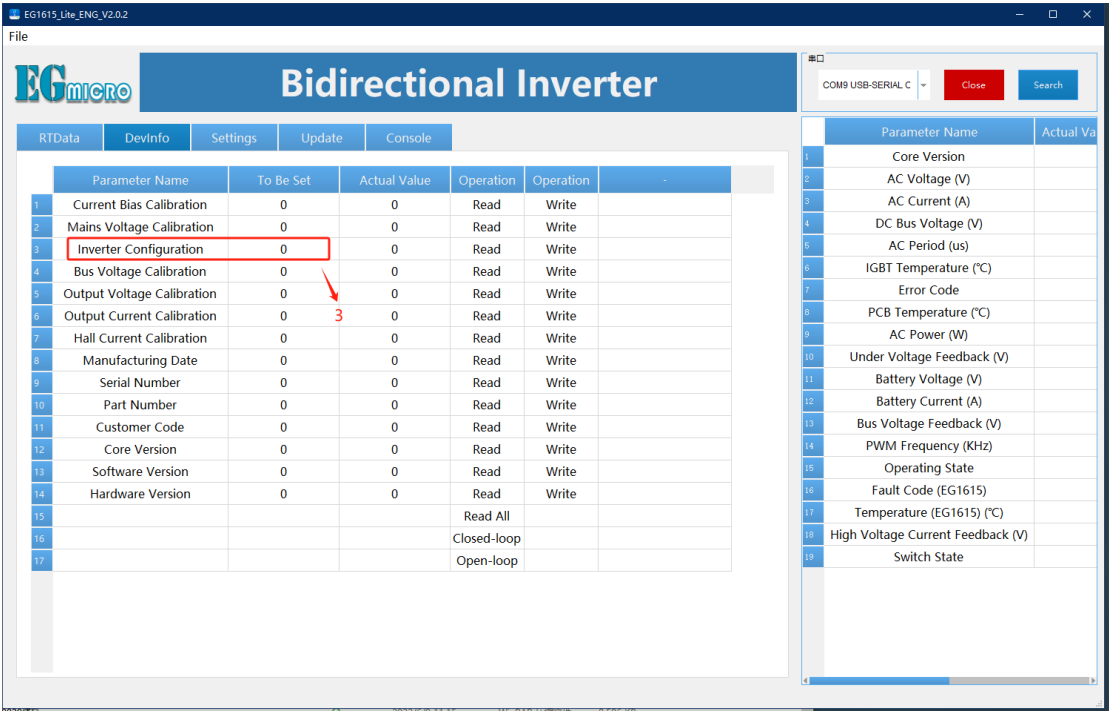


Figure 8-2-2

PS: After performing steps 1 - 4 below for **EG8026**, the inverter control parameters in the advanced settings of EG1615 need to be written as 0 (i.e., custom mode) for the modified parameters to take effect. For **EG8025**, just perform steps 1 - 4 below.

1. **Read CFG configuration:** As shown in Figure 8 - 2 - 1, click "Read All" to read all CFG configurations.
2. **Save all:** As shown in Figure 8 - 2 - 1, after clicking the "Save All" button, save all parameters to the custom area.
3. **Switch to the custom area:** As shown in Figure 8 - 2 - 2, fill in 0 for the inverter configuration to be set, and then click "Write" to switch to the custom area. **PS:** Inverter configuration 0 represents the custom area, and 1 represents the default parameters.
4. **Modify parameters:** As shown in Figure 8 - 2 - 1, click "Write" to write the value to be set in the corresponding register. Ensure the parameters are complete. This operation is not saved, and the settings will be restored after power loss. Only after clicking the "Save All" button will the modified parameters be saved.
5. **Restore defaults:** As shown in Figure 8 - 2 - 1, click the "Restore Defaults" button, and all information will be restored to the factory settings, including the parameters of the device information.

PS: Wait for 5 seconds after restoring defaults and then power on again.

9 Update

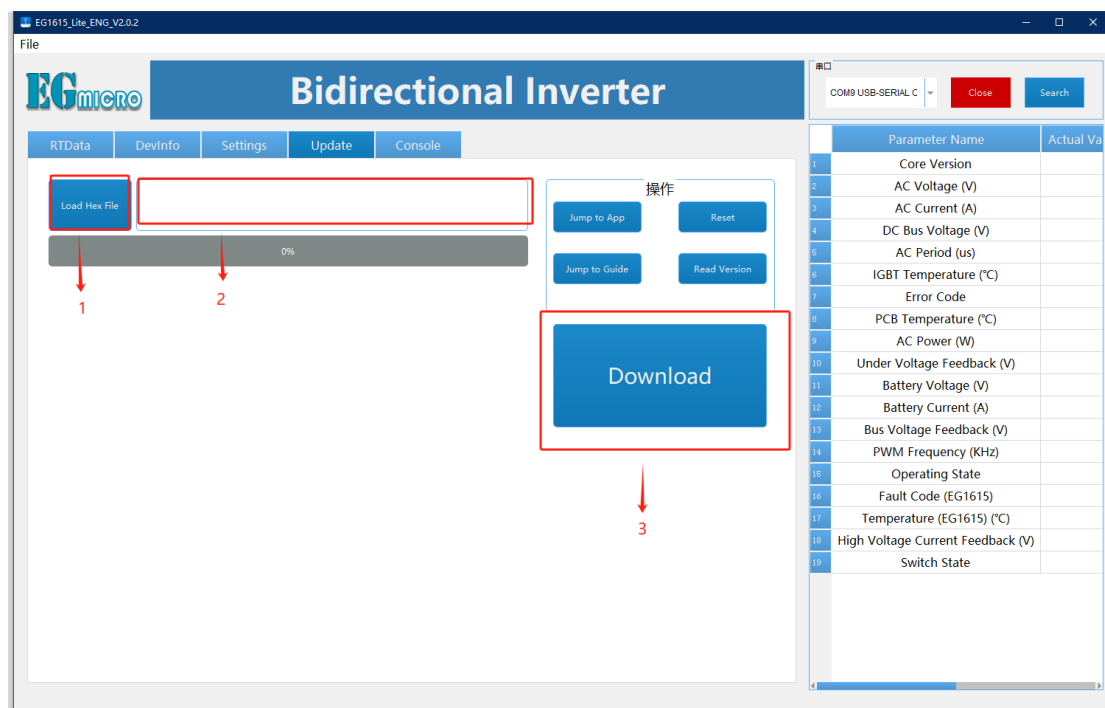


Figure 9-1

1. **Load APP file:** Click "Load APP File" and select the file.
2. **Display file path:** Files can also be directly dragged in.
3. **Update the program:** Click the "Download" button to enter the program download mode.

PS:

- ① In case of a download error, simply click the "Download" button again.
- ② When the whole machine is in operation and the serial port is connected to EG1615, if you need to update the program of EG8026, you need to pass through from EG1615 to EG8026 first, and then click to download. After the download is completed, it needs to be passed back to EG1615.

(For the pass-through function, please refer to the console instructions.)

- ③ If you are updating the driver board program alone, when the serial port is connected to EG1615, directly load the APP file of EG1615. When the serial port is connected to EG8026, directly load the APP file of EG8026.

10 Console

10.1 EG1615 Console

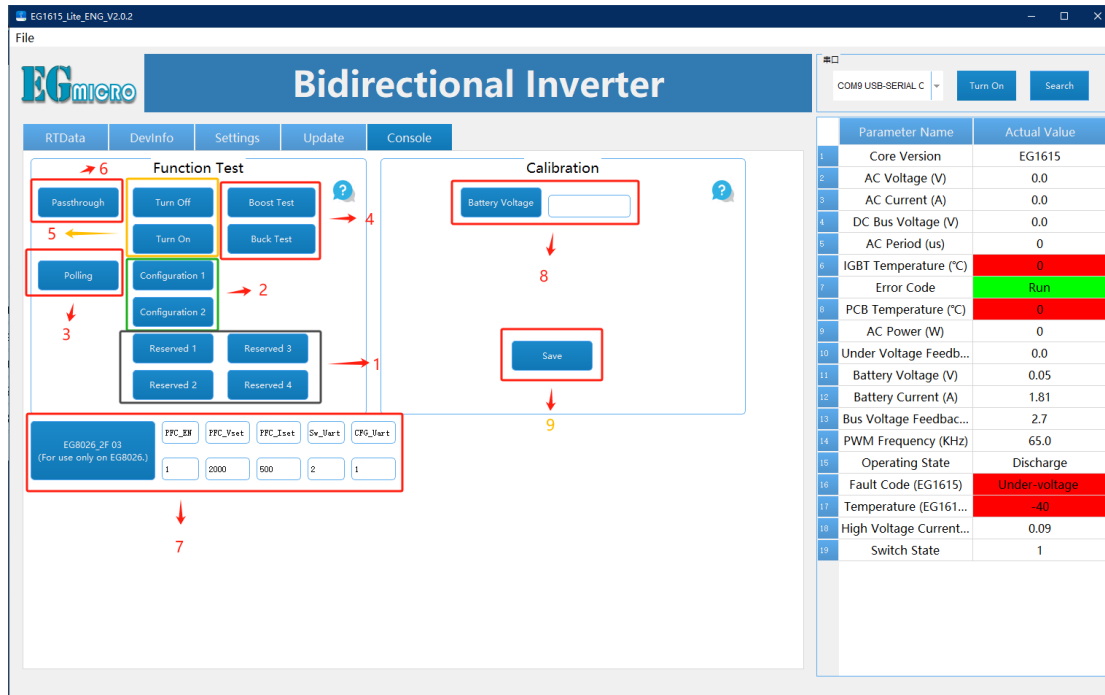


Figure 10-1-1

PS. Instructions for Using When the Serial Port is Connected to EG1615

1. The four buttons are for debugging purposes and have no practical functions.
2. **Configuration 1:** Switch EG8026 to Configuration 1; **Configuration 2:** Switch EG8026 to Configuration 2. The actual output voltage is subject to the configuration of EG8026.
3. **Polling:** When the session is 3, the APP message stops. You can obtain APP data through this button.
4. **Boost Test, Buck Test:** These buttons are for testing purposes only and are used only in low-voltage IO testing. They are prohibited during normal operation.
5. **Turn On:** In the discharging state, it turns on the inverter; in the charging state, it starts charging.
Turn Off: In the discharging state, it turns off the inverter; in the charging state, it stops charging.
6. **Passthrough:** It means receiving and operating the instructions of EG8026 through EG1615.
Specific operation: When the serial port is connected to EG1615, after clicking the "Passthrough" button, you can read the device information and advanced setting parameters of EG8026. At the same time, you can also use functions such as calibration and parameter modification of EG8026. After clicking the "Passthrough" button again, you can switch back to EG1615. During normal operation, the serial port needs to read the data of EG1615. Otherwise, there may be situations where the inverter does not output or the charging does not work.
7. **EG8026_2F_03 Button:** This button is for testing the PFC function of EG8026. It is prohibited during normal operation.
8. **Calibrate Battery Voltage:** During normal operation, measure the actual battery voltage value, multiply it by 100, retain the integer part, fill it into the input box on the right, and then click the " Battery Voltage" button. At this time, the calibration is not saved.
9. **Calibration Save:** After ensuring the calibration is correct, click the "Save" button. At this time, the calibration value is saved.

PS: All calibrations need to be carried out in the inverter state.

10.2•EG8026 Console

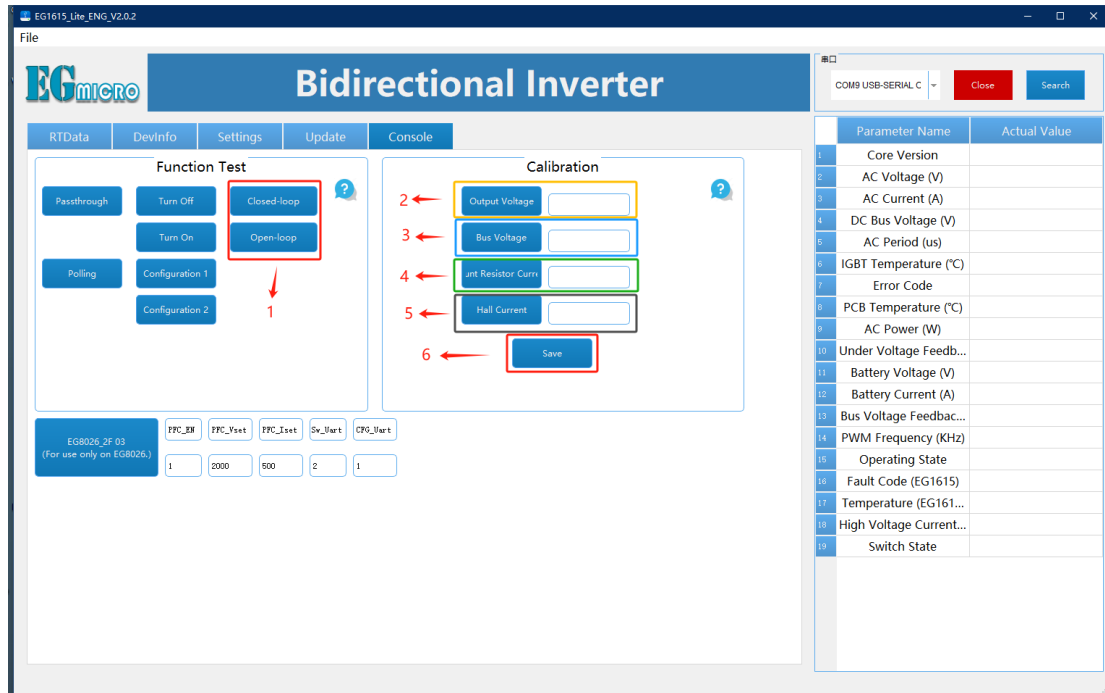


Figure 10-2-1

10.2.1•EG8026 Console Usage Instructions

PS. Instructions for Using When the Serial Port is Connected to EG8026 (All calibrations need to be performed in the inverter state, and the current needs to be calibrated at full load)

1. **Closed-loop**: Controls the closed-loop output of EG8026; **Open-loop**: Controls the open-loop output of EG8026.
2. **Output Voltage** Calibration: During normal operation, measure the actual AC voltage value, multiply it by 10, retain the integer part, fill it into the input box on the right, and then click the "Output Voltage Calibration" button. At this time, the calibration is not saved.
3. **Bus Voltage** Calibration: During normal operation, measure the actual bus voltage value, multiply it by 10, retain the integer part, fill it into the input box on the right, and then click the "Bus Voltage Calibration" button. At this time, the calibration is not saved.
4. **Resistor Current** Calibration: (This calibration value corresponds to the total current of the sampling resistor under the H-bridge arm) During normal operation, measure the actual AC current value, multiply it by 100, retain the integer part, fill it into the input box on the right, and then click the "Output Current Calibration" button. At this time, the calibration is not saved.

5. **Hall Current** Calibration: (This calibration value corresponds to the current collected by the Hall chip) During normal operation, measure the actual AC current value, multiply it by 100, retain the integer part, fill it into the input box on the right, and then click the "Hall Current Calibration" button. At this time, the calibration is not saved.

6. **Save**: After ensuring that the calibrations are correct, click the "Save" button. At this time, all calibration values are saved.

PS. For the instructions of the remaining buttons, please refer to the [EG1615 console instructions](#).

10.2.1•EG8025 Console Usage Instructions

PS2. Instructions for Using When the Serial Port is Connected to EG8025 (All calibrations need to be carried out in the inverter state, and the current needs to be calibrated at full load)

1. **Closed-loop**: Controls the closed-loop output of EG8025; **Open-loop**: Controls the open-loop output of EG8025.
2. **Output Voltage** Calibration: During normal operation, measure the actual AC voltage value, multiply it by 10, retain the integer part, fill it into the input box on the right, and then click the "Output Voltage Calibration" button. At this time, the calibration is not saved.
3. **Bus Voltage** Calibration: During normal operation, measure the actual bus voltage value, multiply it by 10, retain the integer part, fill it into the input box on the right, and then click the "Bus Voltage Calibration" button. At this time, the calibration is not saved.
4. **Resistor Current** Calibration (This calibration value corresponds to the current of the sampling resistor on the left H-bridge arm in EG8025): This calibration method is not recommended for EG8025.
5. **Hall Current** Calibration (This calibration value corresponds to the current of the sampling resistor on the right H-bridge arm in EG8025): This calibration method is not recommended for EG8025.
6. **Save**: After ensuring the calibrations are correct, click the "Save" button. At this time, all calibration values are saved.

PS. For the instructions of the remaining buttons, please refer to the [EG1615 console instructions](#).

11 Q&A

11.1 Q：如何校准电池电压？ How to calibrate the battery voltage?

答：见 [5.1 EG1615 控制台](#)，第 7，8 条。

A: Refer to Article 7 and 8 of the [EG1615 console in Section 10.1](#).

11.2 Q：如何校准 AC 输出电压和输出电流？ How to Calibrate the AC Output Voltage and Output Current?

答：见 [5.2.1 EG8026 控制台](#)

A: Refer to the [EG8026 console in Section 10.2.1](#).

11.3 Q：如何修改参数？ How to modify the parameters?

答：1. EG1615 见 [3.1 EG1615 参数修改](#)。

2. EG8025/EG8026 见 [3.2 EG8026/EG8025 参数修改](#)

A:

1. [For EG1615, refer to 8.1 Parameter Modification of EG1615](#).

2. [For EG8025/EG8026, refer to 8.2 Parameter Modification of EG8026/EG8025](#).

11.4 Q：如何校准电池电流？ How to calibrate the battery current?

答：读出[设备信息](#)中的电池电流校准值后，通过计算写入新的校准值。

当前电池电流校准值*实际电流/显示电流，计算出来的值为新的电池电流校准值。

A: After reading the battery current calibration value in the device information, write in a new calibration value through calculation.

The new battery current calibration value is calculated as the current battery current calibration value multiplied by the actual current and then divided by the displayed current.

11.5 Q: 如何校准 EG8025 的功率或电流? How to calibrate the power or current of EG8025?



图 11-5-1

答：操作步骤见 2~5

1. EG8026 的输出电流校准值对应 EG8025 的左桥电流校准值，EG8026 的霍尔电流校准值对应 EG8025 的右桥电流校准值。
 2. 分别点击 2 中两个“读”按钮，将两个电流校准值读出，读出的值作为当前校准值。
 3. 将上位机 3 中显示的交流电流值作为 A1，实际功率计上显示的电流值为 A2。
 4. 新的校准值计算公式为：新校准值=当前校准值*A2/A1，将新的校准值保留整数后填入 4 中。
 5. 分别写入点击 5 中的“写”，写入后再点击 2 中的“读”，确保写入成功。
- PS：建议将逆变关闭后再执行 5 步骤，逆变关闭可以通过 控制台-关闭。

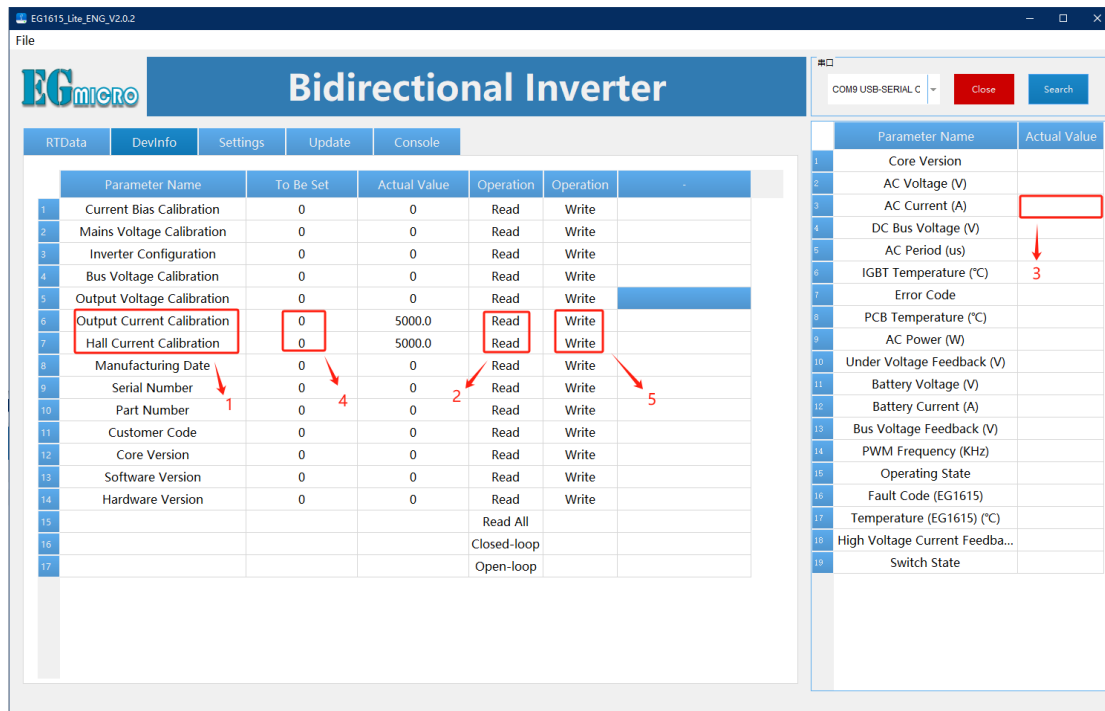


Figure 11-5-2

A: The operation steps are shown in 2 - 5.

1. The output current calibration value of EG8026 corresponds to the left bridge current calibration value of EG8025, and the Hall current calibration value of EG8026 corresponds to the right bridge current calibration value of EG8025.
2. Click the two "Read" buttons in 2 respectively to read out the two current calibration values, and the read values will be used as the current calibration values.
3. Take the AC current value displayed in the host computer 3 as A1, and the current value displayed on the actual power meter as A2.
4. The calculation formula for the new calibration value is: new calibration value = current calibration value * A2 / A1. Fill in the new calibration value after rounding it to an integer in 4.
5. Write respectively by clicking the "Write" in 5. After writing, click the "Read" in 2 again to ensure the writing is successful.

PS: It is recommended to turn off the inverter before performing step 5. The inverter can be turned off through Console – Turn Off.